

Project Progress - Week 1 of March 2025

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This is a file that contains the tracking of the activities related to the OFC project during the first week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the first week of March 2025. The key achievements include:

- Successful implementation and validation of the optical force calculation function in the dipole approximation.
- Development of test and validation scripts that confirm the correct dependence on the permittivity.
- Advancement in the theoretical documentation, particularly in the section on polarizability.
- Detailed derivation of the quasi-static polarizability formula for a dielectric sphere.

2 Potential Tasks for the Week

The following tasks have been identified as potential areas of focus for the first week of March 2025:

- Optimize the functions for querying the refractiveindex.info database.
- Complete the integration of permittivities into an specific equation.
- Finish documenting and providing examples of usage.

3 Week Tracking

3.1 Monday - March 3, 2025

The repository structure has been fixed and the management of Python modules has been addressed. Additionally, I have been learning how to compile LaTeX projects located inside the repository. This document serves as an example of the compilation process.

3.2 Tuesday - March 4, 2025

The tasks for the week have been specified and the progress in documentation, code, and testing has been tracked. We are going to focus on the algorithm used to obtain the polarizability of the particles. In the "dipole momenta" branch, a first version of ".h" and ".cpp" files has been created as a first approach to modular development of the functionalities. I have encountered some issues with warnings and errors flagged by the editor in VSCode. While the code works properly, the editor flags some issues that are not real. It may be related to the editor's settings or installation. Apart from this, it has been observed that a CMakefile must be created, an include directory must be added, and we need to explore how to include different functions without having to import multiple files.

3.3 Thursday - March 6, 2025

The reading of Draine's paper on radiative corrections in polarizability has been initiated. In addition, work has been done on including BibTeX references in LaTeX files. A references file has been created and included in the main file. In this regard, the directory structure of the tracking has been modified. A directory tree has been added to the document to visualize the structure of the tracking directory.

3.4 Friday - March 7, 2025

The difficulties in implementing C++ modules have been discussed with Raúl. He advised me to use include structures, MakeFiles, and temporary build directories. As an example, he recommended me to review his class notes and the structure of his `cpp-template-project` repository. He also advised me to implement Python interfaces that call the C++ code, which performs computationally expensive calculations. In the meeting with Rafa, the need to use the Discrete Dipole Approximation (DDA) for the calculation of optical forces for particles of size comparable to the wavelength has been discussed.

4 Current Next Steps

The next steps for the project include:

- Exploration of CMakefile, include directory, and modular development (Notion).
- Analysis of Raúl's `cpp-template-project` repository.
- Check of Raúl's class notes.
- Development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Understand radiative correction in polarizability. Check the paper [1].
- Documentation of quasi-static polarizability theory.

5 Final State of the Project

This week some milestones have been achieved:

- The repository structure has been fixed.
- The management of Python modules has been addressed.
- The compilation of LaTeX projects located inside the repository has been learned.
- A first version of ".h" and ".cpp" files has been created as a first approach to modular development of the functionalities.
- The reading of Draine's paper on radiative corrections in polarizability has been initiated.
- Work has been done on including BibTeX references in LaTeX files. This has established a good practice for future documentation and tracking structure.
- The difficulties in implementing C++ modules have been discussed with Raúl. And intuition on how to proceed has been provided.
- The need to use the Discrete Dipole Approximation (DDA) for the calculation of optical forces for particles of size comparable to the wavelength has been discussed.

So far, the project is developing in some areas:

- The development of the algorithm to obtain the polarizability of the particles.
- The modular development, including makefile documentation and Raúl's `cpp-template-project` repository and notes analysis.
- The development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.

References

- [1] B. T. Draine. “The Discrete-Dipole Approximation and Its Application to Interstellar Graphite Grains”. In: *The Astrophysical Journal* 333 (Oct. 1988), p. 848. DOI: 10.1086/166795.